

Adrian Domínguez Castro, Ph.D.

Data Scientist & AI/ML Engineer | LLM Systems, RAG & Agentic Workflows | Quantitative Research

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Work Authorization: Permanent Resident (Green Card) / No sponsorship required

SUMMARY:

Quantitative AI/ML Research Engineer (Ph.D. in Theoretical Physics) actively seeking a full-time position to continue building upon my advanced AI/ML skillset and contribute to long-term organizational goals. Leveraging 5+ years of applied machine learning and software engineering experience to architect and deploy frontier GenAI and LLM systems. Specialist in agentic orchestration (LangGraph), context engineering, semantic retrieval, and time-series quantitative modeling. Proven track record designing automated evaluation pipelines, regression suites, and low-latency shared AI services for complex financial, clinical, and scientific datasets.

SELECTED EXPERIENCE

Lead Data & AI/ML Engineer · Sept 2024-Present

Independent AI/ML Consulting

Delivered specialized, project-based technical solutions by partnering directly with executive stakeholders and cross-functional teams to translate complex business problem statements into production-grade AI/ML architectures.

Selected engagements:

Healthcare & research AI: Decimal.health | OncoBrain, AccelerOnc Studio, Moffitt Cancer Center

- Architected end-to-end hybrid RAG systems (Dense + BM25, Cohere Rerank, Vector DBs) for unstructured clinical data. Engineered advanced context assembly and semantic retrieval pipelines (leveraging deep-parsing OCR/extraction tools), achieving high retrieval precision.
- Engineered automated ingestion pipelines using deep-parsing web scrapers and self-healing multi-agent validation loops, reducing data processing latency by 90%.
- Built and orchestrated autonomous AI agents using LangGraph, incorporating deterministic cyclic evaluation, tool routing, and self-correction loops. Packaged these agentic workflows as shared Python services, reducing structured extraction error rates by 30%.

Fintech AI: Stealth Mode Startup

- Engineered end-to-end quantitative financial time-series pipelines (LSTM Autoencoders, RNNs, XGBoost) for market price anomaly detection and signal forecasting, validating models via strict out-of-sample backtesting frameworks. An open-source reference PoC [[WTI Oil Anomaly Detection](#)].
- Designed and executed offline benchmarks and quantitative evaluations comparing SLMs/GenAI against traditional DL models. Built reusable regression suites to make actionable recommendations that optimally balance output quality, compute cost, and inference latency.

Postdoctoral Research Scientist · Feb 2023-Jan 2024

Vanderbilt University, USA

- Architected surrogate ML regression pipelines using traditional ML and deep learning models (Random Forest, XGBoost, ElasticNet, RNNs) on quantum DFT datasets, replacing compute-intensive simulations and optimizing feature extraction workflows.
- Implemented physics-informed ML models to predict material properties, achieving 85% accuracy in benchmarked journal publication [[Paper & Code](#)].
- Engineered high-throughput automated ETL pipelines in Python on HPC infrastructure, parallelizing batch feature extraction and reducing data processing latency by 80%.

Doctoral Research Scientist · Oct 2018-Dec 2022

University of Bremen, Germany

- Led cross-functional research projects across 5 countries, directing end-to-end data processing, stochastic modeling, and simulation architecture for high-dimensional multi-particle dynamic systems.
- Designed scalable data analysis workflows and parallel simulation algorithms in Python and C++, optimizing multi-dimensional array operations and reducing overall processing time by 50%.

- Engineered predictive regression pipelines to model complex non-linear system dynamics across large parameter spaces, implementing automated validation frameworks to ensure numerical reproducibility.
- Published 4 peer-reviewed research papers in top-tier journals, presenting novel computational frameworks and data-driven methodologies to international academic and industry stakeholders.

Associate Research Scientist · Sept 2017-Jul 2018

University of Groningen, The Netherlands

- Developed computational research and mentored junior scientists; built Python automation pipelines for large-scale data analysis (50% processing time reduction) and optimized high-performance simulation code.

EDUCATION

Ph.D. in Theoretical Physics · 2022 · University of Bremen, Germany

B.Sc. in Radiochemistry · 2014 · InSTEC-University of Havana, Cuba

AI SYSTEMS & TECHNICAL EXPERTISE

LLM & GenAI Systems: Fine-tuning, Transformers, Prompt Engineering, RAG, LangChain, LangGraph, Agentic Workflows, Vector Databases (ChromaDB, Qdrant), Model Evaluation, NLP.

Machine Learning & Modeling: Regression (LASSO, Elastic Net), Tree-Based Models (XGBoost, Random Forest), Time Series Analysis & Forecasting, Predictive Modeling, Statistical Analysis.

Deep Learning: Neural Networks (MLP, RNN, LSTM), Physics-Informed ML, Model Evaluation.

Programming & Infrastructure: Python (Production Code, PyTorch, Hugging Face, NumPy, Pandas, scikit-learn), C/C++, SQL, Shell, Linux, HPC, Automated Data Pipelines, Azure ML, GCP

Soft Skills: Executive & Senior Stakeholder Engagement, Cross-Functional Team Leadership, Business-to-Technical Translation, Agile Project Management

SELECTED PUBLICATIONS

- Featured Papers: Author/Co-author of 9 peer-reviewed publications in top-tier scientific journals (including Physical Review Letters, ACS Catalysis, Nanoscale, Theoretical Chemistry Accounts, and Communications Chemistry) focusing on quantum dynamics, density functional theory (DFT), and machine learning applications.
- Full List: Complete publication list and citation metrics available on [Google Scholar Profile](#).